

Akshay Singhal

Postdoctoral Fellow · Department of Civil, Construction, and Environmental Engineering ·
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EDUCATION

Ph.D. in Earth and Environmental Sciences (CPI: 9.2)	2024
Indian Institute of Science Education and Research Bhopal (IISER), India	
Grenoble Alpes University, France	
M.Sc. in Environmental Sciences (CGPA: 8.4; Gold Medalist)	2019
Central University of South Bihar, India	
B.Sc. in Forestry (CPI: 9.32; Silver Medalist)	2017
Sam Higginbottom University of Agriculture, Technology and Sciences, India	

EMPLOYMENT & APPOINTMENTS

Postdoctoral Fellow	Nov 2024 – Present
Department of Civil, Construction, and Environmental Engineering, The University of Alabama, USA	

RESEARCH INTERESTS

- Impact-based Flood Forecasting
- NLP and Deep learning for Disaster Impact Assessment
- Hydrological Extremes and Natural Hazards
- Exposure, Vulnerability, and Resilience Evaluation

PEER-REVIEWED PUBLICATIONS

1. [WRR26] **Singhal, A.**, Karimiziarani, M., & Moradkhani, H. (2026). Comprehensive flood impact assessment using NLP and BERT deep learning for improved impact prediction and disaster management. *Water Resources Research*, 62(4), e2025WR041451. doi: 10.1029/2025WR041451
2. [SERRA26] Samal, N., Meenakshi, K. V., **Singhal, A.**, Jha, S. K., & Oriani, F. (2026). Statistical disaggregation approach for generating hourly precipitation in mountainous regions preserving complex temporal patterns. *Stochastic Environmental Research and Risk Assessment*, 40(3), 61. doi: 10.1007/s00477-026-03179-1
3. [UC25] Prajapati, P., **Singhal, A.**, & Jha, S. K. (2025). Analysis of urban flood vulnerability at the sub-city administrative unit level in Mumbai. *Urban Climate*, 62, 102536. doi: 10.1016/j.uclim.2025.102536
4. [EDS25] **Singhal, A.**, Kavya, M., & Jha, S. K. (2025). A multi-aggregation approach to estimate avalanche vulnerability and suggest phase-wise adaptation. *Environment, Development and Sustainability*, 1–26. doi: 10.1007/s10668-025-06015-8
5. [HESS25] **Singhal, A.**, Crochemore, L., Ruin, I., & Jha, S. K. (2025). INSPIRE Game: Integration of vulnerability in impact-based forecasting of urban floods. *Hydrology and Earth System Sciences*, 29, 947–983. doi: 10.5194/hess-29-947-2025
6. [ATMOS23] **Singhal, A.**, Jaseem, M., & Jha, S. K. (2023). Spatial connections in extreme precipitation events obtained from NWP forecasts: A complex network approach. *Atmospheric Research*, 282, 106538. doi: 10.1016/j.atmosres.2022.106538
7. [WRM23] **Singhal, A.**, Jaseem, M., Sarker, S., Prajapati, P., Singh, A., & Jha, S. K. (2023). Identifying Potential Locations of Hydrologic Monitoring Stations Based on Topographical and Hydrological Information. *Water Resources Management*, 1–16. doi: 10.1007/s11269-023-03675-x
8. [JoH23] **Singhal, A.**, Cheriamparambil, A., Samal, N., & Jha, S. K. (2023). Relating

- forecast and satellite precipitation to generate future skillful ensemble forecasts over the northwest Himalayas at major avalanche and glacier sites. *Journal of Hydrology*, 616, 128795. doi: 10.1016/j.jhydro1.2022.128795
9. [JoHRS23] Samal, N., Ashwin, R., **Singhal, A.**, Jha, S. K., & Robertson, D. E. (2023). Using a Bayesian joint probability approach to improve the skill of medium-range forecasts of the Indian summer monsoon rainfall. *Journal of Hydrology: Regional Studies*, 45, 101284. doi: 10.1016/j.ejrh.2022.101284
 10. [IJRBM23] Singh, A., **Singhal, A.**, Ashwin, R., Samal, N., Pandey, R., & Jha, S. K. (2023). Towards the selection of a longer record of quantitative precipitation forecast dataset from TIGGE ensembles for India during the monsoon season. *International Journal of River Basin Management*, 1–23. doi: 10.1080/15715124.2023.22709711
 11. [EMS22] **Singhal, A.**, Cheriamparambil, A., & Jha, S. K. (2022). Spatial extrapolation of statistically downscaled weather data over the Northwest Himalayas at major glacier sites. *Environmental Modelling & Software*, 149, 105317. doi: 10.1016/j.envsoft.2022.105317
 12. [FES22] **Singhal, A.**, Raman, A., & Jha, S. K. (2022). Potential Use of Extreme Rainfall Forecast and Socio-Economic Data for Impact-Based Forecasting at the District Level in Northern India. *Frontiers in Earth Science*. doi: 10.3389/feart.2022.846113
 13. [IJDRR21] **Singhal, A.**, & Jha, S. K. (2021). Can the approach of vulnerability assessment facilitate identification of suitable adaptation models for risk reduction?. *International Journal of Disaster Risk Reduction*, 63, 102469. doi: 10.1016/j.ijdr.2021.102469
 14. [IJoC21] **Singhal, A.**, & Jha, S. K. (2021). An application of Multiple-point statistics downscaling approach over North-West Himalayas in avalanche-prone areas. *International Journal of Climatology*. doi: 10.1002/joc.7342

BOOK CHAPTER

1. Singhal, A., Prashant (2020). Biofuels: Perspective for Sustainable Development and Climate Change Mitigation. In *Biotechnology for Biofuels: A Sustainable Green Energy Solution*. Springer, Singapore. doi: 10.1007/978-981-15-3761-5_1

CONFERENCE PRESENTATIONS

1. [ORAL] **Singhal, A.**, Karimiziarani, M., Moradkhani, H. (2025). NLP and BERT Deep Learning-based Approaches for Improved Impact-based Forecasting and Disaster Management. *AGU Fall Meeting*.
2. [ORAL] **Singhal, A.**, Karimi, S., Moradkhani, H. (2025). An ML-based approach to develop a metropolitan-scale flood impact database. *HEPEX Workshop*.
3. [ORAL] **Singhal, A.**, Samal, N., Jha, S., Crochemore, L., Ruin, I. (2023). Improved flood-related decision-making in case of urban flash floods in a metropolitan city of India. *EGU General Assembly*. doi: 10.5194/egusphere-egu23-1569
4. [ORAL] Samal, N., **Singhal, A.**, et al. (2023). Statistical generation of fine-resolution precipitation data in Ganga and Godavari River basins of India using limited training datasets. *EGU General Assembly*. doi: 10.5194/egusphere-egu23-1536
5. [ORAL] **Singhal, A.**, Jaseem, M., Singh, A., Jha, S. (2022). Identification of Critical Stream Gauge Locations in the Ganga River Basin. *AGU Fall Meeting*. link: ADS abstract
6. [ORAL] **Singhal, A.**, Raman, A., Jha, S. (2022). Towards use of extreme rainfall forecast and socio-economic data to generate Impact-based forecasts. *EGU General Assembly*. doi: 10.5194/egusphere-egu22-10916
7. [POSTER] **Singhal, A.**, Jha, S. (2021). Assessing the temporal evolution of spatial structure and synchronization of extreme rainfall over Ganga River basin. *AGU Fall*

Meeting. link: ADS abstract

8. [POSTER] Singhal, A., Jha, S. (2021). A Statistical Approach to generate ensembles of observation data of precipitation using hindcast data over the Northwest Himalayas. *EGU General Assembly*. doi: 10.5194/egusphere-egu21-6991
9. [POSTER] Singhal, A., Jha, S. (2021). A Complex Network Approach to Evaluate Multiple Quantitative Precipitation Forecasts Based on Spatial Characteristics of Extreme Rainfall. *INTROMET*.
10. [POSTER] Singhal, A., Jha, S., Samal, N. (2020). A Statistical Approach to extend Fine Scale Weather Data over the Northwest Himalayas. *AGU Fall Meeting*. link: ADS abstract
11. [POSTER] Singhal, A., Jha, S. (2020). A Geostatistical Approach to Generate Fine-Grid Weather Information over the Northwest Himalayas. *TROPMET*.

TEACHING EXPERIENCE

- | | |
|---|----------------------------|
| • Data Analyses and Statistics for Geosciences
Teaching Assistant | IISER Bhopal, Jul–Dec 2021 |
| • Surface Water Hydrology
Teaching Assistant | IISER Bhopal, Jan–Jun 2022 |
| • Contemporary Environmental Issues
Teaching Assistant | IISER Bhopal, Jan–Jun 2022 |
| • Earth Materials and Processes
Teaching Assistant | IISER Bhopal, Jul–Dec 2022 |
| • Climate Change and Extreme Events
Teaching Assistant | IISER Bhopal, Jul–Dec 2023 |

PROFESSIONAL AND COMMUNITY SERVICES

- **Journal Reviewer** for: *Scientific Reports, Journal of Hydrology, Weather and Climate Extremes, Theoretical and Applied Climatology, Modeling Earth Systems and Environment*

INVITED TALKS AND WORKSHOPS

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| • Indian Institute of Tropical Meteorology (IITM), Pune
Seminar: A decision-making experiment for impact-based forecasting of extreme hydro-meteorological events | 2023 |
| • Central University of South Bihar (CUSB), Gaya
Workshop on emergency decision-making during urban flash floods | 2023 |

AWARDS, FELLOWSHIPS, AND GRANTS

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| • Raman–Charpak Award , IFCPAR/CEFIPRA, supporting a six-month research stay in France for work on improved flood-related decision-making using precipitation forecasts and socio-economic data in impact-based forecasting | 2023 |
| • AGU Travel Grant , American Geophysical Union, USA | 2021 |
| • DST-INSPIRE PhD Fellowship , Department of Science and Technology (DST), Government of India | 2020 |
| • Gold Medal (M.Sc.) , Central University of South Bihar | 2019 |
| • UGC-NET (LS) , Qualified | 2018, 2019, 2020 |